

3-72 Attenuation Analysis

In Figure P3-72 a two-port attenuator connects a $600\text{-}\Omega$ source to a $600\text{-}\Omega$ load. Find the power delivered to the load in terms of v_s . Remove the attenuator and find the power delivered to the load when the source is directly connected to the load. By what fraction does the attenuator reduce the power delivered to the $600\text{-}\Omega$ load? Express the fraction in dB.

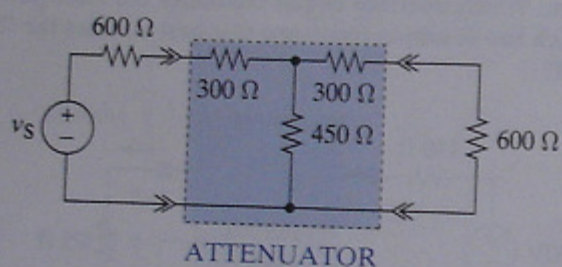


FIGURE P3-72

3-73 Interface Circuit Design

Using no more than three $50\text{-}\Omega$ resistors, design the interface circuit in Figure P3-73 so that $v \leq 4\text{ V}$ and $i \leq 100\text{ mA}$ regardless of the value of R_L .

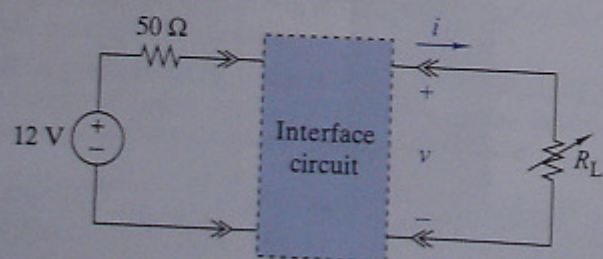
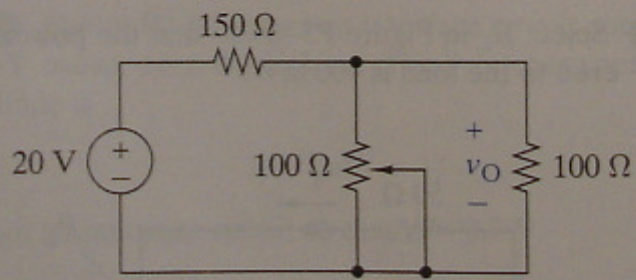


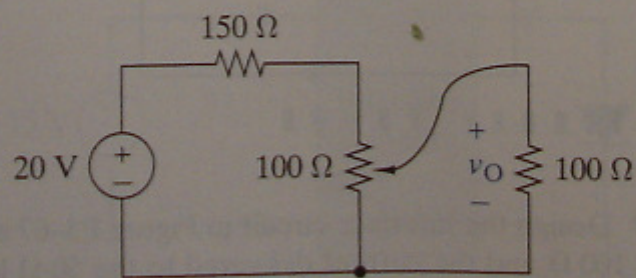
FIGURE P3-73

3-75 E Design Evaluation

A requirement exists for a circuit to deliver 0 to 5 V to a $100\text{-}\Omega$ load from a 20-V source rated at 2.5 W. Two proposed circuits are shown in Figure P3-75. Which one would you choose and why?



Circuit 1



Circuit 2

FIGURE P3-75

- 3-2 (a) Formulate node-voltage equations for the circuit in Figure P3-2.
 (b) Use these equations to find v_x and i_x .

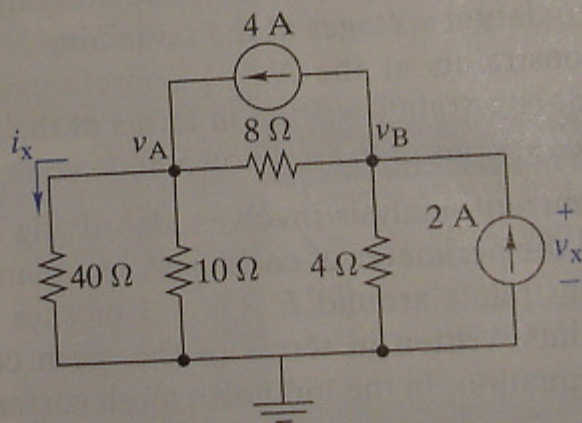


FIGURE P3-2

- 3-5 (a) Formulate node-voltage equations for the circuit in Figure P3-5.
 (b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 10 \text{ k}\Omega$, $v_s = 20 \text{ V}$, and $i_s = 2 \text{ mA}$.

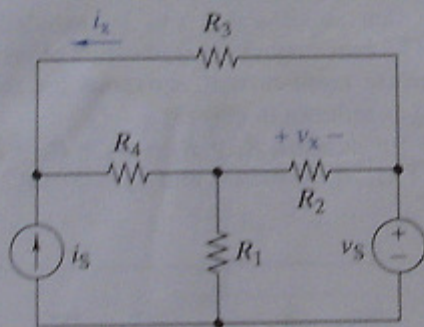


FIGURE P3-5

3-6 (a) Formulate node-voltage equations for the circuit in Figure P3-6.

(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = R_5 = 10\text{ k}\Omega$, and $v_s = 20\text{ V}$.

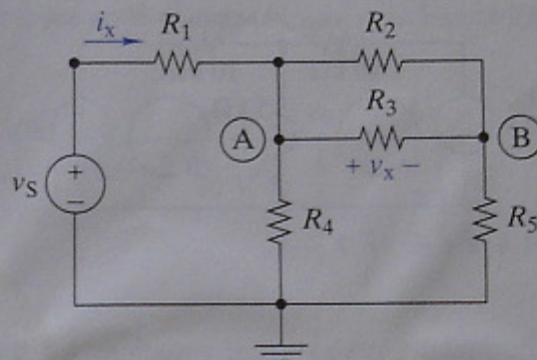


FIGURE P3-6

3-7 (a) Formulate node-voltage equations for the circuit in Figure P3-7.

(b) Solve for v_A , v_B , and v_C when $R_1 = 1\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$, $R_3 = 4\text{ k}\Omega$, $R_4 = 2\text{ k}\Omega$, and $i_{s1} = i_{s2} = 2\text{ mA}$.

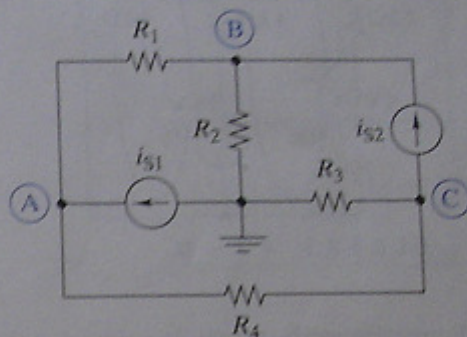


FIGURE P3-7

3-9 (a) Formulate mesh-current equations for the circuit in Figure P3-9.

(b) Use these equations to find v_x and i_x .

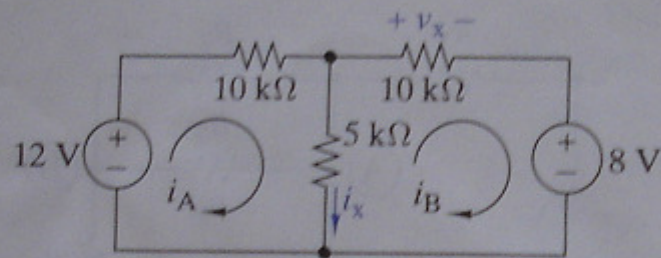


FIGURE P3-9

- 3-16 (a) Formulate mesh-current equations for the circuit in Figure P3-16.
 (b) Use these equations to find the input resistance.

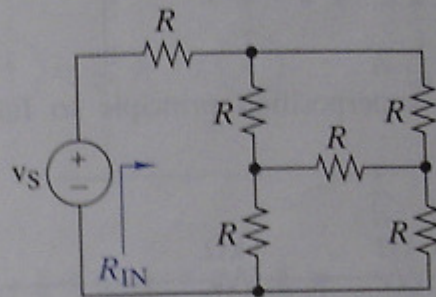


FIGURE P3-16

- 3-17 In Figure P3-17 all of the resistors are $1\text{ k}\Omega$. The voltage at node C is found to be $v_C = -2\text{ V}$ when node B is connected to ground. Find the node voltages v_A and v_D , and the mesh currents i_A and i_B .

